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Lab 2

- 1-create two workspaces dev and prod
- 2-create two variable definition files(.tfvars) for the two environments
- 3-separate network resources into network module
- 4-apply your code to create two environments one in us-east-1 and eu-central-1
- 5-run local-exec provisioner to print the public_ip of bastion ec2
- 6- upload infrastructure code on github project
- 10-create rds (mysql)
- 11-create elastic cache (redis)

Amazon S3 > Buckets > terraform-lab2-state > env:/

Amazon S3 **env:/** [Copy S3 URI](#)

Buckets
General purpose buckets
Directory buckets
Table buckets
Vector buckets

Files
File systems [New](#)

Access management and security
Access Points
Access Points for FSx

Objects (2)
Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

Name	Type	Last modified	Size	Storage class
dev/	Folder	-	-	-
prod/	Folder	-	-	-

Tables (1) [Info](#)

Find tables

Filter by tag: Any tag key

Filter by tag value: Any tag value

Last updated: May 3, 2026, 16:46 (UTC+3:00)

[Actions](#) [Delete](#) [Create table](#)

Name	Status	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode	Write capacity mode	Total size	Table class
terraform-lock	Active	LockID [S]	-	0	0	Off	☆	On-demand	On-demand	0 bytes	Standard

Your VPCs
[VPCs](#) [VPC encryption controls](#)

Your VPCs (2) [Info](#)

Find VPCs by attribute or tag

Last updated: 16 minutes ago

[Actions](#) [Create VPC](#)

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv4 CIDR	IPv6 CIDR
-	vpc-0e82a81de3c727423	Available	-	-	Off	10.0.0.0/16	-
-	vpc-050c617a1a38ab784	Available	-	-	Off	172.31.0.0/16	-

Subnets (9) [Info](#)

Find subnets by attribute or tag

Last updated: 16 minutes ago

[Actions](#) [Create subnet](#)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR
-	subnet-041b4ac3e2db55f51	Available	vpc-0e82a81de3c727423	Off	10.0.2.0/24	-
-	subnet-0701b6809253db45c	Available	vpc-0e82a81de3c727423	Off	10.0.3.0/24	-
-	subnet-019b3f4ea94f291dd	Available	vpc-050c617a1a38ab784	Off	172.31.80.0/20	-
-	subnet-0d41f895de4ce5e67	Available	vpc-050c617a1a38ab784	Off	172.31.32.0/20	-
-	subnet-0235f5e698a855a64	Available	vpc-0e82a81de3c727423	Off	10.0.1.0/24	-
-	subnet-0a2b4fd5f8e816f0c	Available	vpc-050c617a1a38ab784	Off	172.31.48.0/20	-
-	subnet-0f459081e537b011c	Available	vpc-050c617a1a38ab784	Off	172.31.16.0/20	-
-	subnet-083abaa65418292c	Available	vpc-050c617a1a38ab784	Off	172.31.64.0/20	-
-	subnet-0d59387128a04ca03	Available	vpc-050c617a1a38ab784	Off	172.31.0.0/20	-

Route tables (3) [Info](#)

Find route tables by attribute or tag

Last updated: 16 minutes ago

[Actions](#) [Create route table](#)

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
-	rtb-0fd27d46e38d17600	-	-	Yes	vpc-0e82a81de3c727423	325614625863
-	rtb-07570621fca094a8f	subnet-0235f5e698a855...	-	No	vpc-0e82a81de3c727423	325614625863
-	rtb-0bd23220b234c5fe3	-	-	Yes	vpc-050c617a1a38ab784	325614625863

Security Groups (4) Info

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner
-	sg-05c6df8b246ff5bb8	default	vpc-0e82a81de3c727423	default VPC security group	325614625863
-	sg-0ef9e8ae8caf5737	default	vpc-050c617a1a38ab784	default VPC security group	325614625863
-	sg-0ba6972e8dd47b1504	redis-sg-default	vpc-0e82a81de3c727423	Managed by Terraform	325614625863
-	sg-0220b229da61ad6bf	rds-security-group	vpc-0e82a81de3c727423	Managed by Terraform	325614625863

Instances (1) Info

Last updated less than a minute ago

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic IP
-	i-017e142c79acbe22b	Running	t3.micro	3/3 checks pass	View alarms	us-east-1c	-	54.235.11.80	-

Redis OSS caches (1) Info

Upgrade to Valkey View details Actions Create cache

Find Redis caches

Cache name	Status	Description	Engine version	Configuration	Created date
dev-redis	Available	-	7.1.0	cache.t3.micro	May 3, 2026, 17:58:31 (UTC+03:00)

Databases (1)

Group resources Modify Actions Create database

Filter by databases

DB identifier	Status	Role	Engine	Upgrade rollout ...	Region ...	Size	Recommendations
terraform-202605031453455997000000	Available	Instance	MySQL Community	SECOND	us-east-1a	db.t3.micro	

7-verify your email in ses service

8- create lambda function to send email

9-create trigger to detect changes in state file and send the email

Amazon SES Configuration: Identities mustafatarek104@gmail.com

Announcing availability of email validation in SES. Use email validation to help you protect your sender reputation. Learn more

mustafatarek104@gmail.com Delete Send test email

Summary

Identity status: Verified

Amazon Resource Name (ARN): arn:aws:ses:us-east-1:325614625863:identity/mustafatarek104@gmail.com

AWS Region: US East (N. Virginia)

Code source Info

Open in Visual Studio Code Upload from

lambda_function.py

```

1 import boto3
2
3 def lambda_handler(event, context):
4     ses = boto3.client('ses')
5
6     # Get details about the change from the S3 event
7     bucket = event['Records'][0]['s3']['bucket']['name']
8     key = event['Records'][0]['s3']['object']['key']
9
10    message = f"Alert: The Terraform state file was updated\nBucket: {bucket}\nPath: {key}"
11
12    ses.send_email(
13        Source='mustafatarek104@gmail.com',
14        Destination={'ToAddresses': ['mustafatarek104@gmail.com']},
15        Message={
16            'Subject': {'Data': 'Terraform State Change Detected'},
17            'Body': {'Text': {'Data': message}}
18        }
19    )
20    return {"status": "Email Sent"}
```

Amazon EventBridge

Event notifications (1)

Send a notification when specific events occur in your bucket. Learn more

Name	Event types	Filters	Destination type	Destination
StateChangeTrigger	All object create events	env/	Lambda function	TerraformStateNotifier

Amazon EventBridge

For additional capabilities, use Amazon EventBridge to build event-driven applications at scale using S3 event notifications. Learn more or see EventBridge pricing

Send notifications to Amazon EventBridge for all events in this bucket

Off

TerraformStateNotifier

Throttle

Copy ARN

Actions ▾

▼ Function overview Info

Diagram

Template

Export to Infrastructure Composer

Download ▾



+ Add destination

+ Add trigger

Function ARN

arn:aws:lambda:us-east-1:325614625863:function:TerraformStateNotifier

Description

-

Last modified

51 minutes ago

Terraform State Change Detected

Inbox ×



mustafatarek104@gmail.com via [amazon](#)ses.com
to me ▾

Alert: The Terraform state file was updated!
Bucket: env:/dev/terraform.tfstate